

KAGGLE COMPETITION

Sofia ML Regression 2026

Understanding the dataset & the modeling approach

A walk through what the data looks like, the challenge it poses, and the sequence of approaches that follow from it.



THE DATASET

What we're working with

2,500

training rows

15

anonymous features (x0 – x14)

1

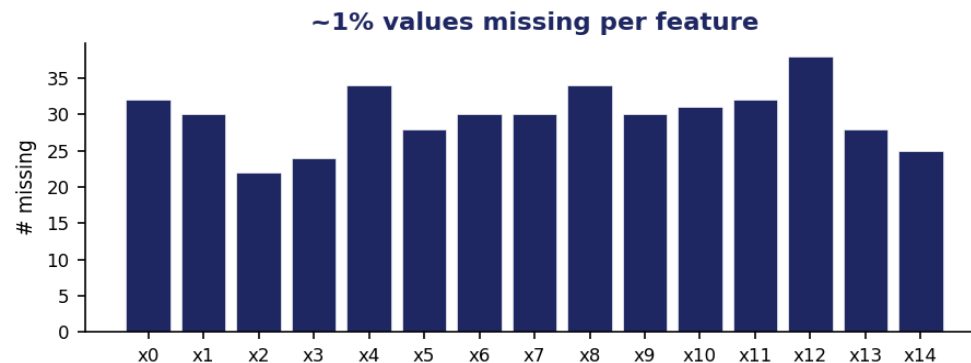
continuous target to predict

~1%

values missing per feature (448 total)

NATURE OF THE DATA

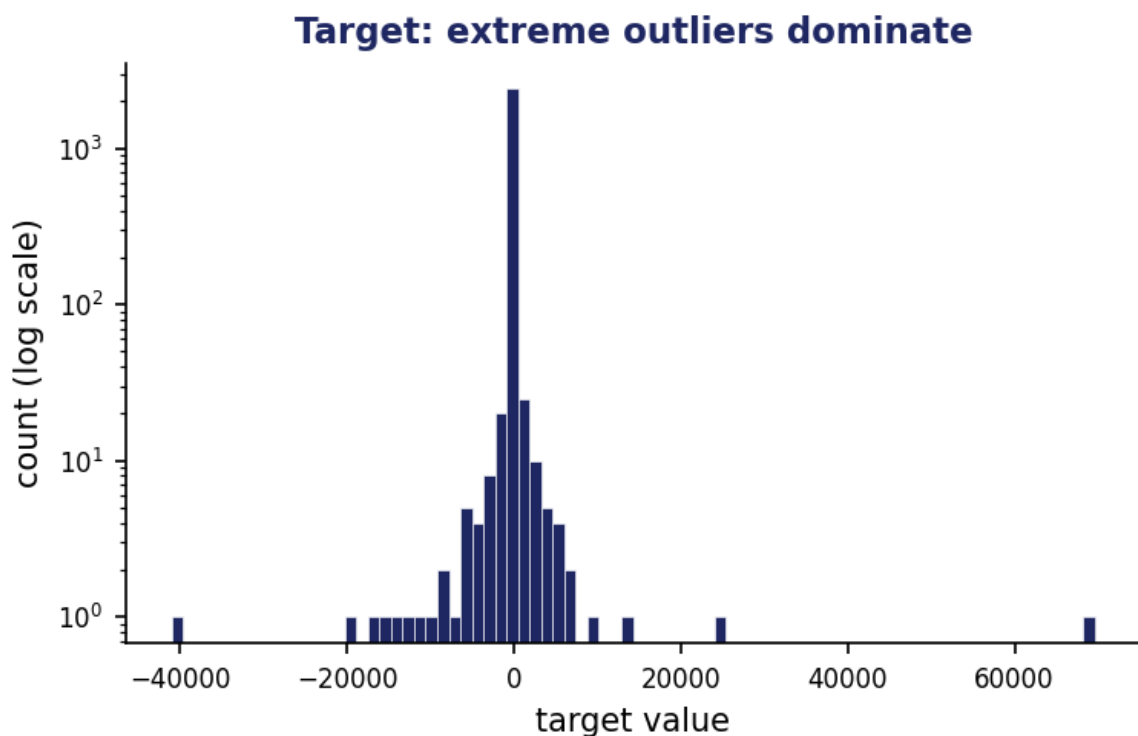
- All 15 features are continuous and unlabeled — no domain meaning given.
- A pure regression problem: map 15 numbers to one.
- Missing values are sparse and spread evenly across every column.





DATASET — KEY FINDING 1

The target is ruled by rare extremes



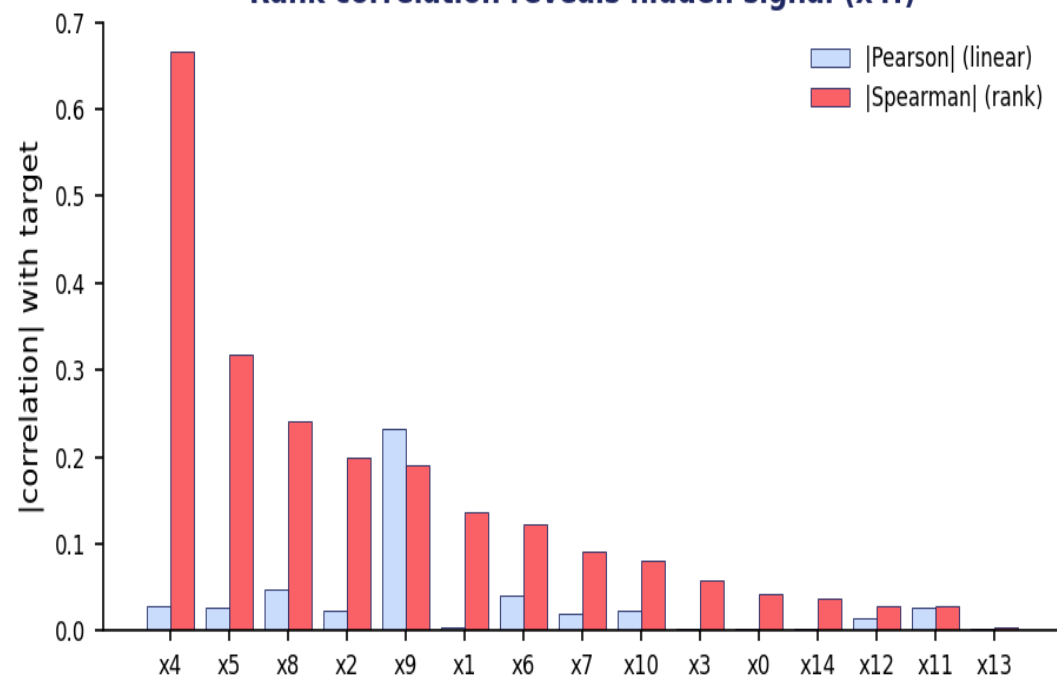
WHAT THIS MEANS

- Most targets sit in a narrow band, but a small ~4.5% of rows reach as far as $\pm 70,000$.
- Those few extreme rows carry almost all of the target's variance.
- Any squared-error metric becomes hyper-sensitive to how those rows are handled.
- Models that chase the extremes risk huge errors; models that ignore them miss most of the variance.

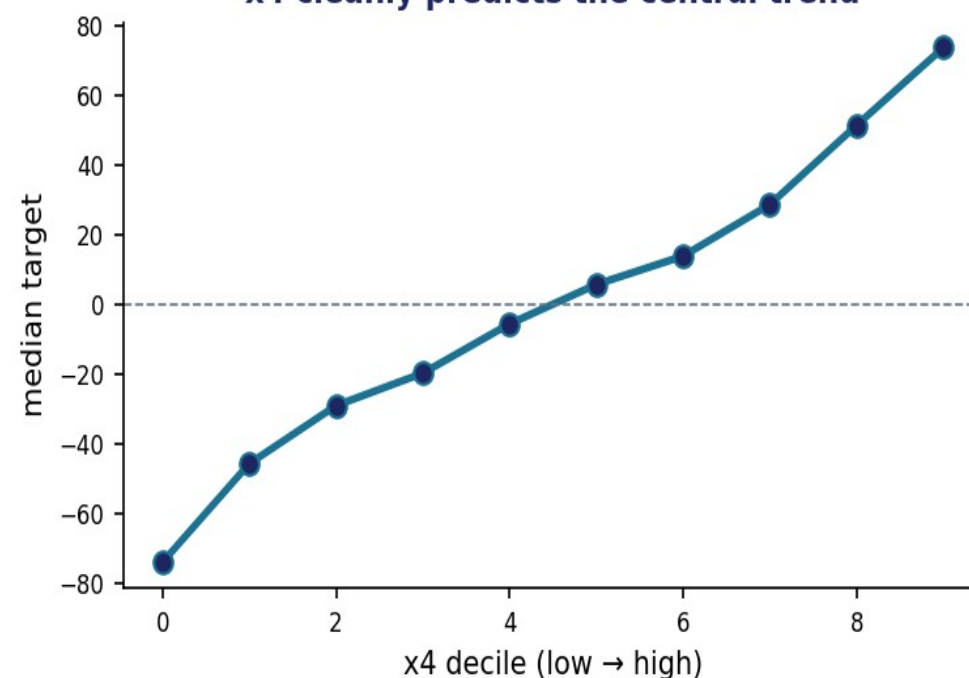


The real signal is monotonic, not linear

Rank correlation reveals hidden signal (x4!)



x4 cleanly predicts the central trend



Left: x4 shows almost no *linear* correlation, yet a strong *rank* correlation (+0.67) — the outliers hide it from least-squares. **Right:** median target climbs smoothly across x4 — a clean monotonic relationship to exploit with rank/quantile features.



Each approach answered the last one's flaw

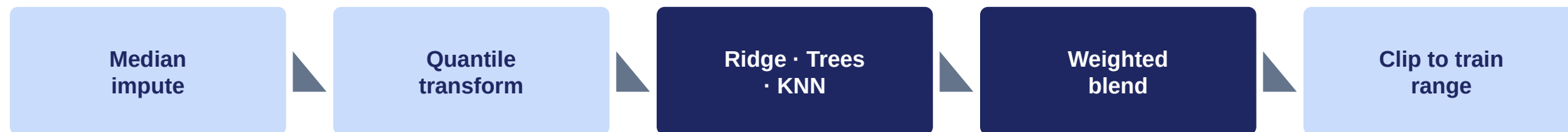
Approach	Motivation — why we tried it	What it taught us
Polynomial of x_9 (x_9^3)	x_9^3 tracks the extreme rows almost perfectly in training	Unbounded cubes extrapolate wildly on unseen values
Plain linear regression	Mirror the simplest possible baseline	Still unbounded — large features → large risky predictions
Quantile-transform + Ridge	Rank-normalize features to expose the monotonic signal and bound the inputs	Captures the real signal safely, but modest on its own
Outlier-robust regression	Down-weight extremes (Huber / winsorize) to fit the central trend	Stable, but leaves the high-variance extremes unmodeled
Bounded ensemble (final)	Average diverse bounded models so independent errors cancel	Robust and well-behaved on out-of-range data

The throughline: *move from fragile, unbounded fits toward bounded, diverse models — robustness over raw flexibility.*



A bounded, diverse ensemble

PIPELINE



WHY BOUNDED

- Quantile-transformed inputs live in a fixed range, and outputs are clipped to the training target's span.
- So a test row with an unusual feature value can never be amplified into a runaway prediction.

WHY DIVERSE

- A linear model, trees, and a nearest-neighbour model make largely uncorrelated errors.
- Averaging them cancels independent mistakes, which is exactly what a noisy, outlier-heavy target needs.



Code: https://github.com/klok-828/ml-hw-spring-2026-python/blob/main/challenge_v11.ipynb